Possible Problems for Quiz 5 to be taken 2/14/19. Of these problems I will choose 3 to be on the quiz.

Problem 1

Find the inverse of $A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

Problem 2

Solve the linear system $x_1 + x_2 - x_3 = 1, \quad x_2 + 2x_3 = 2, \quad x_3 = 3$ using your result from Problem 1.

Problem 3

Determine if $S = \{\vec{u} \in R^2 \mid u_1 \; and \; u_2 \; are \; rational \; numbers \}$ is a subspace.

Problem 4

Show that $B = \left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$ forms a basis for R^2. Why is $S = \left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix}\right\}$ not a basis?

Problem 5

Consider the set of all positions of a penny of mass m_p placed at a point x_1 and dime of mass m_d placed at a point x_2 on a 10 inch ruler such that the center of mass is in the middle. If you place those positions into a vector $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, justify why the set of all possible $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ comes from a subspace.

Problem 6

Prove the following Theorem

> Let $B = \{\vec{u}_1, \vec{u}_2, \cdots, \vec{u}_m\}$ be a basis for a subspace S. For every vector $\vec{s} \in$ S there exists a unique set of scalars $s_1, s_2 \; \ldots, s_m$ such that the linear combination
>
> $$s_1\vec{u}_1 + s_2\vec{u}_2 + \cdots + s_m\vec{u}_m = \vec{s}$$

Problem 7

Go here: https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

What is the quote at the beginning of the second video in the list, the video on span.

Also, I highly recommend this set of youtube videos. They will really lend some intuition to what we are studying this semester.